

17

THE DIFFRACTION GRATING

Any arrangement which is equivalent in its action to a number of parallel equidistant slits of the same width is called a *diffraction grating*. Since the grating is a very powerful instrument for the study of spectra, we shall treat in considerable detail the intensity pattern which it produces. We shall find that the pattern is quite complex in general but that it has a number of features in common with that of the double slit

17.1 EFFECT OF INCREASING THE NUMBER OF SLITS

When the intensity pattern due to one, two, three, and more slits of the same width is photographed, a series of pictures like those shown in Fig. 17A(a) to (f) is obtained.

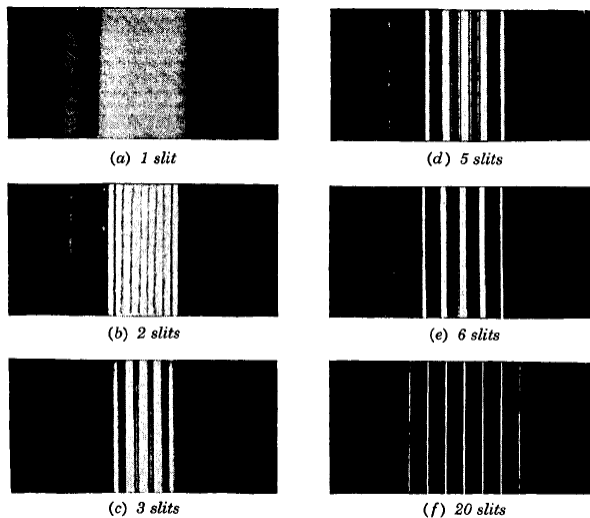


FIGURE 17A
Fraunhofer diffraction patterns for gratings containing different numbers of slits.

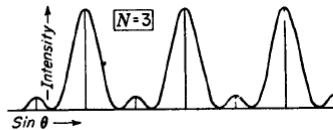
The arrangement of light source, slit, lenses, and recording plate used in taking these pictures was similar to that described in previous chapters, and the light used was the blue line from a mercury arc. These patterns therefore are produced by *Fraunhofer diffraction*. In fact, it was because of Fraunhofer's original investigations of the diffraction of parallel light by gratings in 1819 that his name became associated with this type of diffraction.

The most striking modification in the pattern as the number of slits is increased consists of a narrowing of the interference maxima. For two slits these are diffuse, having an intensity which was shown in the last chapter to vary essentially as the square of the cosine. With more slits the sharpness of these *principal maxima* increases rapidly, and in pattern (f) of the figure, with 20 slits, they have become narrow lines.

Another change, of less importance, which can be seen in patterns (c), (d), and (e) is the appearance of weak *secondary maxima* between the principal maxima, their number increasing with the number of slits. For three slits only one secondary maximum is present, its intensity being 11.1 percent of the principal maximum. Figure 17B shows an intensity curve for this case, plotted according to the theoretical equation

$$A^2 = a^2 \frac{\sin^2 (N\delta/2)}{\sin^2 (\delta/2)} = a^2 \frac{\sin^2 N\gamma}{\sin^2 \gamma} \quad (17b)$$

FIGURE 17B
Principal and secondary maxima from a
grating of three slits.



17.2 INTENSITY DISTRIBUTION FROM AN IDEAL GRATING

The procedure used in Secs. 15.2 and 16.2 for the single and double slits could be used here, performing the integration over the clear aperture of the slits, but it becomes cumbersome. Instead let us apply the more powerful method of adding the complex amplitudes

We designate this magnitude by a and the number of slits by N .

The phase will change by equal amounts δ from one slit to the next; so the resultant complex amplitude is the sum of the series

$$Ae^{i\theta} = a(1 + e^{i\delta} + e^{i2\delta} + e^{i3\delta} + \dots + e^{i(N-1)\delta}) = a \frac{1 - e^{iN\delta}}{1 - e^{i\delta}} \quad (17a)$$

To find the intensity, this expression must be multiplied by its complex conjugate, as in Eq. (14m), giving

$$A^2 = a^2 \frac{(1 - e^{iN\delta})(1 - e^{-iN\delta})}{(1 - e^{i\delta})(1 - e^{-i\delta})} = a^2 \frac{1 - \cos N\delta}{1 - \cos \delta}$$

Using the trigonometric relation $1 - \cos \alpha = 2 \sin^2 (\alpha/2)$, we may then write

$$A^2 = a^2 \frac{\sin^2 (N\delta/2)}{\sin^2 (\delta/2)} = a^2 \frac{\sin^2 N\gamma}{\sin^2 \gamma} \quad (17b)$$

where, as in the double slit, $\gamma = \delta/2 = (\pi d \sin \theta)/\lambda$.

the factor a^2 represents the intensity diffracted by a single slit, and after inserting its value from Eq. (15d) we finally obtain for the intensity in the Fraunhofer pattern of an ideal grating

$$\bullet \quad I \approx A^2 = A_0^2 \frac{\sin^2 \beta}{\beta^2} \frac{\sin^2 N\gamma}{\sin^2 \gamma} \quad (17c)$$

Upon substitution of $N = 2$ in this formula, it readily reduces to Eq. (16c) for the double slit.

17.3 PRINCIPAL MAXIMA

The new factor $(\sin^2 N\gamma)/(\sin^2 \gamma)$ may be said to represent the *interference* term for N slits. It possesses maximum values equal to N^2 for $\gamma = 0, \pi, 2\pi, \dots$. Although the quotient becomes indeterminate at these values, this result can be obtained by noting that

$$\lim_{\gamma \rightarrow m\pi} \frac{\sin N\gamma}{\sin \gamma} = \lim_{\gamma \rightarrow m\pi} \frac{N \cos N\gamma}{\cos \gamma} = \pm N$$

These maxima correspond in position to those of the double slit, since for the above values of γ

$$d \sin \theta = 0, \lambda, 2\lambda, 3\lambda, \dots = m\lambda \quad \text{Principal maxima} \quad (17d)$$

The relative intensities of the different orders m are in all cases governed by the single-slit diffraction envelope $(\sin^2 \beta)/\beta^2$. Hence the relation between β and γ in terms of slit width and slit separation [Eq. (16d)] remains unchanged, as does the condition for missing orders [Eq. (16h)].

17.4 MINIMA AND SECONDARY MAXIMA

To find the minima of the function $(\sin^2 N\gamma)/(\sin^2 \gamma)$, we note that the numerator becomes zero more often than the denominator, and this occurs at the values $N\gamma = 0, \pi, 2\pi, \dots$ or, in general, $p\pi$. In the special cases when $p = 0, N, 2N, \dots$, γ will be $0, \pi, 2\pi, \dots$; so for these values the denominator will also vanish, and we have the principal maxima described above. The other values of p give zero intensity, since for these the denominator does not vanish at the same time. Hence the condition for a minimum is $\gamma = p\pi/N$, excluding those values of p for which $p = mN$, m being the order. These values of γ correspond to path differences

$$d \sin \theta = \frac{\lambda}{N}, \frac{2\lambda}{N}, \frac{3\lambda}{N}, \dots, \frac{(N-1)\lambda}{N}, \frac{(N+1)\lambda}{N}, \dots \quad \text{Minima} \quad (17e)$$

omitting the values $0, N\lambda/N, 2N\lambda/N, \dots$, for which $d \sin \theta = m\lambda$ and which according to Eq. (17d) represent principal maxima. Between two adjacent principal maxima there will hence be $N - 1$ points of zero intensity. The two minima on either side of a principal maximum are separated by twice the distance of the others.

Between the other minima the intensity rises again, but the secondary maxima thus produced are of much smaller intensity than the principal maxima. Figure 17C shows a plot for six slits of the quantities $\sin^2 N\gamma$ and $\sin^2 \gamma$, and also of their quotient, which gives the intensity distribution in the interference pattern.

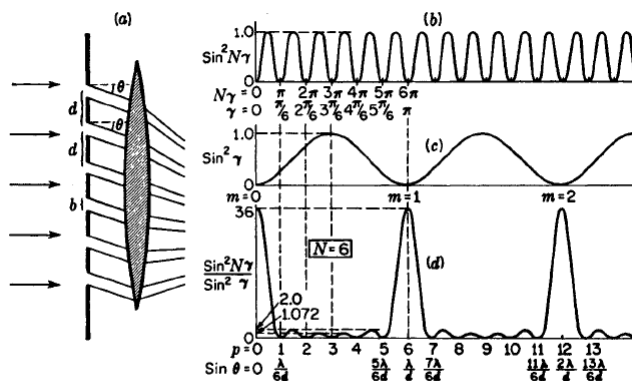


FIGURE 17C
Fraunhofer diffraction by a grating of six very narrow slits and details of the intensity pattern.

The intensity of the principal maxima is N^2 or 36, so that the lower figure is drawn to a smaller scale.

The intensities of the secondary maxima are also shown. These secondary maxima are not of equal intensity but fall off as we go out on either side of each principal maximum. Nor are they in general equally spaced, the lack of equality being due to the fact that the maxima are not quite symmetrical. This lack of symmetry is greatest for the secondary maxima immediately adjacent to the principal maxima,

As the number of slits is increased, the number of secondary maxima is also increased, since it is equal to $N - 2$.

17.5 FORMATION OF SPECTRA BY A GRATING

The secondary maxima discussed in Sec. 17.4 are of little importance in the production of spectra by a many-lined grating. The principal maxima treated in Sec. 17.3 are called *spectrum lines* because when the primary source of light is a narrow slit they become sharp, bright lines on the screen. These lines will be parallel to the rulings of the grating if the slit also has this direction. For monochromatic light of wavelength λ , the angles θ at which these lines are formed are given by Eq. (17d), which is the ordinary grating equation $d \sin \theta = m\lambda$ commonly given in elementary textbooks. A more general equation includes the possibility of light incident on the grating at any angle i . The equation then becomes

$$\bullet \quad d(\sin i + \sin \theta) = m\lambda \quad \text{Grating equation} \quad (17f)$$

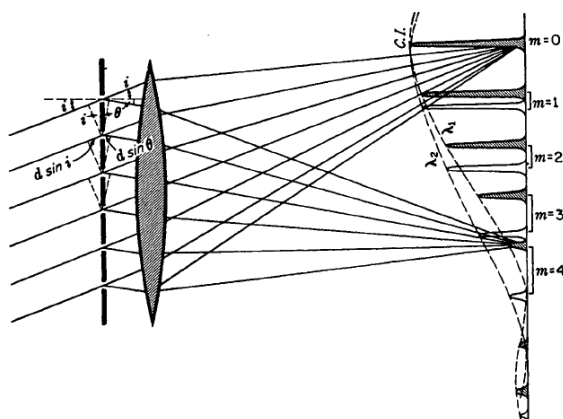


FIGURE 17E
Positions and intensities of the principal maxima from a grating where light containing two wavelengths is incident at an angle i and diffracted at various angles θ .

since, as will be seen from Fig. 17E, this is the path difference for light passing through adjacent slits. The figure shows the path of the light forming the maxima of order $m = 0$ (called the *central image*), and also $m = 4$ in light of a particular wavelength λ_1 . For the central image, Eq. (17f) shows that $\sin \theta = -\sin i$, or $\theta = -i$. The negative sign comes from the fact that we have chosen to call i and θ positive when measured on the same side of the normal; i.e., our convention of signs is such that whenever the rays used *cross over* the line normal to the grating, θ is taken as negative.

Those intensity maxima which are shaded show the various orders of the wavelength λ_1 . In the case of the fourth order, for example, the path differences indicated are such that $d(\sin i + \sin \theta) = 4\lambda_1$. The intensities of the principal maxima are limited by the diffraction pattern corresponding to a single slit (broken line) and drop to zero at the first minimum of that pattern, which here coincides with the fifth order. The missing orders in this illustration are therefore $m = 5, 10, \dots$, as would be produced by having $d = 5b$.

Now if the source gives light of another wavelength λ_2 somewhat greater than λ_1 , the maxima of the corresponding order m for this wavelength will, according to Eq. (17j), occur at larger angles θ . Since the spectrum lines are narrow, these maxima will in general be entirely separate in each order from those of λ_1 and we have two lines forming a *line spectrum* in each order. These spectra are indicated by brackets in the figure. Both the wavelengths will coincide, however, for the central image, because for this the path difference is zero for any wavelength. A similar set of spectra occurs on the other side of the central image, the shorter wavelength line in each order lying on the side toward the central image. Figure 17F shows actual photographs of grating spectra corresponding to the diagram of Fig. 17E. If the source gives white light, the central image will be white, but for the orders each will be spread out into *continuous spectra* composed of an infinite number of adjacent images of the slit in light of the different wavelengths present. At any given point in such a continuous spectrum, the light will be very nearly monochromatic because of the narrowness of the slit images formed by the grating and lens. The result is in this respect fundamentally different from that with the double slit, where the images were broad and the spectral colors were not separated.

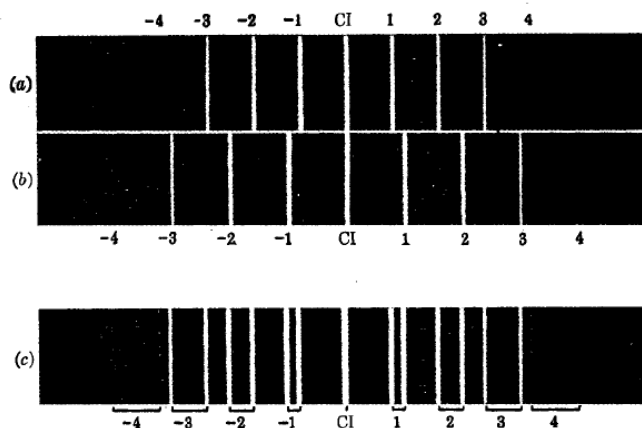


FIGURE 17F
Grating spectra of two wavelengths: (a) $\lambda_1 = 4000 \text{ \AA}$; (b) $\lambda_2 = 5000 \text{ \AA}$; (c) λ_1 and λ_2 together.

17.6 DISPERSION

The separation of any two colors, such as λ_1 and λ_2 in Figs. 17E and 17F, increases with the order number. To express this separation the quantity frequently used is called the *angular dispersion*, which is defined as the rate of change of angle with change of wavelength. An expression for this quantity is obtained by differentiating Eq. (17f) with respect to λ , remembering that i is a constant independent of wavelength. Substituting the ratio of finite increments for the derivative, one has

$$\bullet \quad \frac{\Delta\theta}{\Delta\lambda} = \frac{m}{d \cos \theta} \quad \text{Angular dispersion} \quad (17g)$$

The equation shows in the first place that for a given small wavelength difference $\Delta\lambda$, the angular separation $\Delta\theta$ is directly proportional to the order m . Hence the second-order spectrum is twice as wide as the first order, the third three times as wide as the first, etc. In the second place, $\Delta\theta$ is inversely proportional to the slit separation d , which is usually referred to as the *grating space*. The smaller the grating space, the more widely spread the spectra will be. In the third place, the occurrence of $\cos \theta$ in the denominator means that in a given order m the dispersion will be smallest on the normal, where $\theta = 0$, and will increase slowly as we go out on either side of this.

If θ does not become large, $\cos \theta$ will not differ much from unity, and this factor will be of little importance. If we neglect its influence, the different spectral lines in one order will differ in angle by amounts which are directly proportional to their difference in wavelength. Such a spectrum is called a *normal spectrum*, and one of the chief advantages of gratings over prism instruments is this simple linear scale for wavelengths in their spectra.

17.8 WIDTH OF THE PRINCIPAL MAXIMA

It was shown at the beginning of Sec. 17.4 that the first minima on either side of any principal maximum occur where $N\gamma = mN\pi \pm \pi$, or where $\gamma = m\pi \pm (\pi/N)$. When $\gamma = m\pi$, we have the principal maxima, owing to the fact that the phase difference δ or 2γ , in the light from corresponding points of adjacent slits, is given by $2\pi m$, or a whole number of complete vibrations. However, if we change the angle enough to cause a change of $2\pi/N$ in the phase difference, reinforcement no longer occurs, but the light from the various slits now interferes to produce zero intensity. A phase difference of $2\pi/N$ between the maximum and the first minimum means a path difference of λ/N .

To see why this path difference causes zero intensity, consider Fig. 17G(a), in which the rays leaving the grating at the angle θ form a principal maximum of order m . For these, the path difference of the rays from two adjacent slits is $m\lambda$, so that all the waves arrive in phase. The path difference of the *extreme* rays is then $Nm\lambda$, since N is always a very large number in any practical case.* Now let us change the angle of diffraction by a small amount $\Delta\theta$, such that the extreme path difference increases by one wavelength and becomes $Nm\lambda + \lambda$ (rays shown by broken lines). This should correspond to the condition for zero intensity, because, as required,

the path difference for two adjacent slits has been increased by λ/N . It will be seen that the ray from the top of the grating is now of opposite phase from that at the center, and the effects of these two will cancel. Similarly, the ray from the next slit below the center will annul that from the next slit below the top, etc. The cancellation if continued will yield zero intensity from the whole grating, in entire analogy to the similar process considered in Sec. 15.3 for the single-slit pattern.

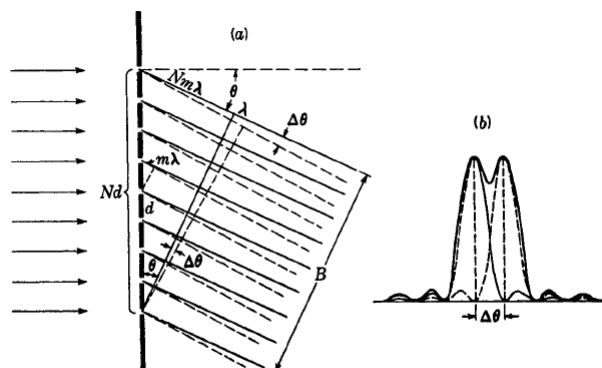


FIGURE 17G
Angular separation of two spectrum lines which are just resolved by a diffraction grating.

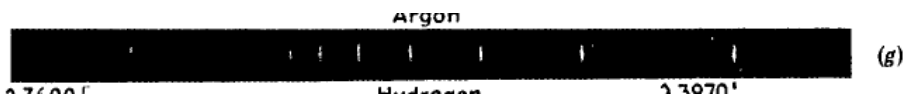
Thus the first zero occurs at the small angle $\Delta\theta$ on each side of any principal maximum. From the figure it is seen that

$$\bullet \quad \Delta\theta = \frac{\lambda}{B} = \frac{\lambda}{Nd \cos \theta} \quad \text{Angular half width of principal maximum} \quad (17i)$$

It is instructive to note that this is just $1/N$ th of the separation of adjacent orders, since the latter is represented by the same expression with the path difference $N\lambda$ instead of λ in the numerator.

17.12 GHOSTS

In an actual grating the ruled lines will always deviate to some extent from the ideal of equal spacing. This gives rise to various effects, according to the nature of the ruling error. Three types can be distinguished. (1) The error is *perfectly random* in magnitude and direction. In this case the grating will give a continuous spread of light underlying the principal maxima, even when monochromatic light is used. (2) The error *continuously increases* in one direction. This can be shown to give the grating "focal properties." Parallel light after diffraction is no longer parallel but slightly divergent or convergent. (3) The error is *periodic* across the surface of the grating. This is the most common type, since it frequently arises from defects in the driving mechanism of the ruling machine. It gives rise to "ghosts," or false lines, accompanying every principal maximum of the ideal grating. When there is only one period involved in the error, these lines are symmetrical in spacing and intensity about the principal maxima. Such ghosts are called *Rowland ghosts*, and may easily be seen in Fig. 21H(g).



(g) Balmer series of hydrogen in the ultraviolet, $\lambda = 3600$ to $4000 \text{ \AA}$. This is a grating spectrum. The faint lines on either side of the stronger members are false lines called ghosts.

17.14 MEASUREMENT OF WAVELENGTH WITH THE GRATING

Small gratings 2 to 5 cm wide are usually mounted on the prism table of a small spectrometer with collimator and telescope. By measuring the angles of incidence and diffraction for a given spectrum line its wavelength can be calculated from the grating formula [Eq. (17f)]. For this the grating space d must be known, and this is usually furnished with the grating. The first accurate wavelengths were determined by this method, the grating space being found by counting the lines in a given distance with a traveling microscope. Once the absolute wavelength of a single line is known, others can be measured relative to it by using the overlapping of orders. For instance, according to Eq. (17h), a sodium line of wavelength $5890 \text{ \AA}$ in the third order will coincide with another line of $\lambda = \frac{2}{3} \times 5890 = 4417 \text{ \AA}$ in the fourth order. Of course no two lines will exactly coincide in this way, but they may fall close enough together to permit the small difference to be accurately corrected for. This method of comparing wavelengths is not accurate with the arrangement described above because the telescope lens is never perfectly achromatic and the two lines will not be focused in exactly the same plane. To avoid this difficulty Rowland invented the *concave grating*, in which the focusing is done by a concave mirror, upon which the grating itself is ruled.

17.15 CONCAVE GRATING

If the grating, instead of being ruled on a plane surface, is ruled on a concave spherical mirror of metal, it will diffract and focus the light at the same time, thus doing away with the need for lenses. Besides the fact that this eliminates the chromatic aberration mentioned above, it has the great advantage that the grating can be used for regions of the spectrum which are not transmitted by glass lenses, such as the ultraviolet.

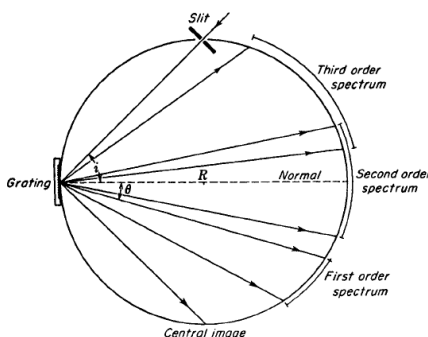


FIGURE 17M
Paschen mounting for a concave grating.